TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	59.0 ± 0.00
Correlation Stability	70.0 ± 0.00
Distribution Consistency	74.2 ± 0.87
Entropy Preservation	33.2 ± 0.98
Mean Z Score Drift	0.5 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	27.8 ± 3.43
Data Density	100.0 ± 0.00
Coverage Score	37.3 ± 2.53
Feature Accuracy	38.8 ± 0.40
Novelty Score	0.6 ± 0.02

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Dataset Audit: chronic_kidney_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	400
Columns	25
Numeric features	25
Categorical features	0
Missing cells (%)	10.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	51.483	45.911	-5.573	10.8%	Warn
bp	76.469	72.923	-3.546	4.6%	OK
sg	1.017	1.022	+0.005	0.4%	OK
al	1.017	0.000	-1.017	100.0%	Drift
su	0.450	0.000	-0.450	100.0%	Drift
rbc	0.190	0.000	-0.190	100.0%	Drift
pc	0.227	0.000	-0.227	100.0%	Drift
pcc	0.106	0.000	-0.106	100.0%	Drift
ba	0.056	0.000	-0.056	100.0%	Drift
bgr	148.037	109.245	-38.791	26.2%	Drift
bu	57.426	33.338	-24.088	41.9%	Drift
sc	3.072	0.974	-2.098	68.3%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	59.0 ± 0.00
Correlation Stability (*2)	70.0 ± 0.00
Distribution Consistency (*3)	74.2 ± 0.87
Entropy Preservation (*4)	33.2 ± 0.98
Mean Z Score Drift (*5)	0.5 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	27.8 ± 3.43

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	37.3 ± 2.53
Feature Accuracy (*10)	38.8 ± 0.40
Novelty Score (*11)	0.6 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6024999999999999	N/A	0.5562499999999999	-0.046250000000000006	
F1	0.7507214882964476	N/A	0.5431461023825549	-0.20757538591389285	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9825000000000002	N/A	0.9862500000000001	0.003749999999999977	
F1	0.9848619694663338	N/A	0.9883155452642038	0.003453575797870043	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.99875	N/A	0.9974999999999999	-0.001249999999999955	
F1	0.9989010989010989	N/A	0.997893803347744	-0.0010072955533547488	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6125	N/A	0.55	-0.062499999999999986	
F1	0.5885198185573248	N/A	0.4349133991438408	-0.15360641941348407	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.735	N/A	0.70875	-0.026250000000000016	
F1	0.6975438258366131	N/A	0.6747429907588897	-0.022800835077723448	

Model: DT

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9425000000000001	N/A	0.9412499999999999	-0.00124999999999999955	
F1	0.9425936579576868	N/A	0.9459136807903052	0.00332002283261850 77	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9862500000000001	N/A	0.9887500000000001	0.00250000000000000 22	
F1	0.9881288971593788	N/A	0.9905442296806999	0.00241533252132117 86	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9525	N/A	0.95375	0.00125000000000000 67	
F1	0.9587663012688215	N/A	0.9597798135439561	0.00101351227513450 94	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9100000000000001	N/A	0.9200000000000002	0.01000000000000002	
F1	0.92384446063128	N/A	0.9313833092820343	0.00753884865075429 3	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.97875	N/A	0.97875	-1.1102230246251566e-17	
F1	0.9805471229310466	N/A	0.9807958251916444	0.00024870226059786 615	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	48.0	80.0	1.02	1.0	0.0	nan	0.0	0.0	0.0	121.0
2	7.0	50.0	1.02	4.0	0.0	nan	0.0	0.0	0.0	nan
3	62.0	80.0	1.01	2.0	3.0	0.0	0.0	0.0	0.0	423.0
4	48.0	70.0	1.005	4.0	0.0	0.0	1.0	1.0	0.0	117.0
5	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
6	60.0	90.0	1.015	3.0	0.0	nan	nan	0.0	0.0	74.0
7	68.0	70.0	1.01	0.0	0.0	nan	0.0	0.0	0.0	100.0
8	24.0	nan	1.015	2.0	4.0	0.0	1.0	0.0	0.0	410.0
9	52.0	100.0	1.015	3.0	0.0	0.0	1.0	1.0	0.0	138.0
10	53.0	90.0	1.02	2.0	0.0	1.0	1.0	1.0	0.0	70.0
11	50.0	60.0	1.01	2.0	4.0	nan	1.0	1.0	0.0	490.0
12	63.0	70.0	1.01	3.0	0.0	1.0	1.0	1.0	0.0	380.0
13	68.0	70.0	1.015	3.0	1.0	nan	0.0	1.0	0.0	208.0
14	68.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	98.0
15	68.0	80.0	1.01	3.0	2.0	0.0	1.0	1.0	1.0	157.0
16	40.0	80.0	1.015	3.0	0.0	nan	0.0	0.0	0.0	76.0
17	47.0	70.0	1.015	2.0	0.0	nan	0.0	0.0	0.0	99.0
18	47.0	80.0	nan	nan	nan	nan	nan	0.0	0.0	114.0
19	60.0	100.0	1.025	0.0	3.0	nan	0.0	0.0	0.0	263.0
20	62.0	60.0	1.015	1.0	0.0	nan	1.0	1.0	0.0	100.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	35.861617161067535	73.1026952684459	1.025	0.0	0.0	0.0	0.0	0.0	0.0	127.58485148320261
2	46.920776195371545	67.56601983719038	1.025	0.0	0.0	0.0	0.0	0.0	0.0	101.73592065123849
3	32.414127232149223	74.11895957341623	1.0220594797867082	0.0	0.0	0.0	0.0	0.0	0.0	85.11672829860863
4	52.167642893319005	70.0	1.0237732837351832	0.0	0.0	0.0	0.0	0.0	0.0	112.14901590517249
5	43.252229388207134	77.47770611792863	1.025	0.0	0.0	0.0	0.0	0.0	0.0	127.98216489434249
6	61.197800282186655	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	106.11941340858311
7	35.0	80.0	1.005	3.0	0.0	1.0	0.0	0.0	0.0	nan
8	30.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	96.0
9	52.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	99.0
10	65.0	80.0	1.015	0.0	0.0	nan	0.0	0.0	0.0	115.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	50.056086620781616	73.30996390800766	1.025	0.0	0.0	0.0	0.0	0.0	0.0	98.92288089642528
2	46.525814185283844	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	105.57275896438638
3	43.951868722827337	78.09430676877703	1.02	0.0	0.0	0.0	0.0	0.0	0.0	114.52259879941013
4	74.393932454553556	67.97977484851188	1.02	0.0	0.0	0.0	0.0	0.0	0.0	107.40404503029762
5	31.955220075257211	72.68656370477927	1.02	0.0	0.0	0.0	0.0	0.0	0.0	124.85074903617658
6	54.936978955068364	60.88585087077638	1.025	0.0	0.0	0.0	0.0	0.0	0.0	128.45659651689448
7	64.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	113.0
8	42.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	98.0
9	12.0	60.0	1.015	3.0	0.0	1.0	1.0	1.0	0.0	nan
10	23.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	48.7981352650037	80.0	1.02060093236749	0.0	0.0	0.0	0.0	0.0	0.0	102.168018360483
	5		8							88
2	49.6774423585571	76.7744235855713	1.02	0.0	0.0	0.0	0.0	0.0	0.0	132.968029481964
	4	8								24
3	36.0982659144872	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	91.4393063657949
	7									
4	30.2815186938687	63.9520478271360	1.02302397608643	0.0	0.0	0.0	0.0	0.0	0.0	115.742457392563
	58	3	2							22
5	42.6418278005837	78.8060926686124	1.02470152316715	0.0	0.0	0.0	0.0	0.0	0.0	82.6714702639425
	4	6	3							5
6	43.2636730080816	60.0	1.02133440249711	0.0	0.0	0.0	0.0	0.0	0.0	121.787811532903
	8		36							99
7	34.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.0
8	33.0	60.0	nan	nan	nan	0.0	0.0	0.0	0.0	130.0
9	60.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	124.0
10	57.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	132.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	37.9745223145984	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	112.534235675091
										2
2	51.4387495684198	62.7742184802624	1.02069355462006	0.0	0.0	0.0	0.0	0.0	0.0	133.393515975658
	65	2	55							86
3	43.4285807805939	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	85.6072433913854
	9									2
4	56.2888034908075	75.7039883639747	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.956751128694
	74	6								
5	60.3723254714958	67.8102712262465	1.02	0.0	0.0	0.0	0.0	0.0	0.0	116.503566038005
	5	4								54
6	46.7602998571717	68.3585172906304	1.025	0.0	0.0	0.0	0.0	0.0	0.0	82.8791170049738
	3	3								9
7	44.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0
8	55.0	90.0	1.01	2.0	1.0	1.0	1.0	0.0	0.0	273.0
9	45.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	360.0
10	72.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	308.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	66.44216685608168	72.03757931998055	1.0210187896599903	0.0	0.0	0.0	0.0	0.0	0.0	103.33235722403501
2	39.09360047348147	60.02925014796296	1.0200146250739814	0.0	0.0	0.0	0.0	0.0	0.0	85.97952489642593
3	40.92592238022671	80.0	1.0200529125855524	0.0	0.0	0.0	0.0	0.0	0.0	121.98941748288954
4	32.9294784161802	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	103.13728267340842
5	55.90851465392037	76.97161782026544	1.025	0.0	0.0	0.0	0.0	0.0	0.0	117.28079484511483
6	23.2153581929529	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	99.08973258039704
7	41.0	80.0	1.015	1.0	4.0	1.0	0.0	0.0	0.0	210.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	87.0
9	26.0	70.0	1.015	0.0	4.0	nan	0.0	0.0	0.0	250.0
10	50.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	137.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	45.235847508713333	80.0	1.0224599768952596	0.0	0.0	0.0	0.0	0.0	0.0	73.93596303241536
2	57.581029150086766	71.8994282499518	1.025	0.0	0.0	0.0	0.0	0.0	0.0	112.59222982518799
3	54.700880852943411	70.0	1.0212134317118762	0.0	0.0	0.0	0.0	0.0	0.0	95.29911914705659
4	27.793023000743247	60.0	1.0247930230007432	0.0	0.0	0.0	0.0	0.0	0.0	79.66232639762161
5	29.595686420662553	77.45650617242612	1.025	0.0	0.0	0.0	0.0	0.0	0.0	118.64126543106528
6	40.512003362001809	65.7599974138447	1.0228799987069224	0.0	0.0	0.0	0.0	0.0	0.0	94.7200077584659
7	8.0	60.0	1.025	3.0	0.0	0.0	0.0	0.0	0.0	78.0
8	60.0	70.0	1.01	1.0	0.0	nan	0.0	0.0	0.0	109.0
9	nan	70.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	110.0
10	62.0	90.0	1.02	2.0	1.0	nan	0.0	0.0	0.0	169.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	25.37329436622118	70.0	1.020051846439753	0.0	0.0	0.0	0.0	0.0	0.0	88.33181721441883
2	50.54267842712183	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	100.95419404751014
3	48.86833981706643	60.0	1.022028474923776	0.0	0.0	0.0	0.0	0.0	0.0	110.01071443595484
4	39.54486084903352	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	133.86345825471005
5	43.6088900391513	67.34321115461256	1.025	0.0	0.0	0.0	0.0	0.0	0.0	102.92246772992618
6	37.28511461535414	63.72276200465277	1.021861381002323	0.0	0.0	0.0	0.0	0.0	0.0	107.45943778557641
7	nan	70.0	1.02	0.0	0.0	nan	nan	0.0	0.0	219.0
8	29.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.0
9	50.0	90.0	1.015	1.0	0.0	1.0	1.0	0.0	0.0	nan
10	65.0	70.0	1.02	1.0	0.0	1.0	1.0	0.0	0.0	139.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	43.27028162380938	62.70281623809384	1.025	0.0	0.0	0.0	0.0	0.0	0.0	110.2429594047654
2	43.97959334260768	61.85528483162240	1.020463821207905	0.0	0.0	0.0	0.0	0.0	0.0	114.64934969106784
3	51.53313908271461	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	125.32680264209978
4	50.20141242337651	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	92.67137474458838
5	42.26745227014658	80.0	1.022589441434558	0.0	0.0	0.0	0.0	0.0	0.0	102.83813338426333
6	43.12459094376292	79.75081811247415	1.025	0.0	0.0	0.0	0.0	0.0	0.0	129.82557267873199
7	53.0	100.0	1.01	1.0	3.0	1.0	0.0	0.0	0.0	213.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	nan
9	45.0	80.0	1.02	3.0	0.0	0.0	1.0	0.0	0.0	425.0
10	33.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	89.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	53.89868713649743	73.15485604884813	1.025	0.0	0.0	0.0	0.0	0.0	0.0	86.10708637069112
2	40.34346512639158	68.88044572800702	1.0244402228640035	0.0	0.0	0.0	0.0	0.0	0.0	124.55977713599655
3	44.0	63.25897201716984	1.0216294860085848	0.0	0.0	0.0	0.0	0.0	0.0	94.02230839484905
4	34.00718792722312	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	100.98402682839307
5	53.594001764474285	75.16916593146905	1.02	0.0	0.0	0.0	0.0	0.0	0.0	113.64283053958238
6	41.80263412983966	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	115.99560978360022
7	46.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	102.0
8	22.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	97.0
9	65.0	80.0	1.01	3.0	3.0	nan	0.0	0.0	0.0	294.0
10	23.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	47.0662243979832	62.33443900504201	1.0193395448366518	0.0	0.0	0.0	0.0	0.0	0.0	110.5993365969748
2	60.0	72.71100497232499	1.0213555024861625	0.0	0.0	0.0	0.0	0.0	0.0	123.71100497232499
3	43.94197370455589	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	94.57678948182236
4	53.995350012241246	65.24030951798036	1.0185174213285428	0.0	0.0	0.0	0.0	0.0	0.0	107.42790714460509
5	41.666379170326245	73.88936804945627	1.0230553159752718	0.0	0.0	0.0	0.0	0.0	0.0	94.72255763461938
6	52.565515899736377	76.38475357866986	1.025	0.0	0.0	0.0	0.0	0.0	0.0	133.9386550101378
7	11.0	80.0	1.01	3.0	0.0	nan	0.0	0.0	0.0	nan
8	61.0	90.0	1.015	0.0	2.0	nan	0.0	0.0	0.0	nan
9	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
10	70.0	60.0	nan	nan	nan	nan	nan	0.0	0.0	120.0